task_2 .py - C:/Users/josth/AppData/Local/Programs/Python/Python313/task_2 .py (3.13.17)

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```
from decimal import Decimal
import sys

# Explore data types
print("DATA TYPES")
print(type(10))
print(type(10.0))
print(type("10"))
print(type(True))
print(type([10, 8.5]))

# Division operators
print("\nDIVISION OPERATORS")
print("7 / 2 =", 7 / 2)
print("7 // 2 =", 7 // 2)
print("7 % 2 =", 7 % 2)
print("-7 // 2 =", -7 // 2)

# Floating-point issue
print("\nFLOATING POINT ISSUE")
print("0.1 + 0.2 =", 0.1 + 0.2)
print("0.1 + 0.2 == 0.3:", 0.1 + 0.2 == 0.3)
print("Decimal result:", Decimal("0.1") + Decimal("0.2"))
print("Decimal equality:",
      Decimal("0.1") + Decimal("0.2") == Decimal("0.3"))

# Payment calculations
payments = ["33.40"] * 1000 + ["0.10"] * 3


def total_float(items):
    total = 0.0
    for amount in items:
        total += float(amount)
    return total


def total_decimal(items):
    total = Decimal("0.00")
    for amount in items:
        total += Decimal(amount)
    return total
```

```python
def total_rounded(items):
    total = 0.0
    for amount in items:
        total = round(total + float(amount), 2)
    return total


print("\nPAYMENT TOTALS")
print("float  :", total_float(payments))
print("Decimal:", total_decimal(payments))
print("Paise  :", total_paise(payments), "paise")
print("Rupees :", total_paise(payments) / 100)
print("Rounded:", total_rounded(payments))

# Decimal from float comparison
print("\nDECIMAL CONVERSION")
print("Decimal(amount):",
      sum((Decimal(x) for x in payments), Decimal("0.00")))
print("Decimal(float(amount)):",
      sum((Decimal(float(x)) for x in payments), Decimal("0.00")))

# Memory analysis
print("\nMEMORY USAGE")
print("Integer:", sys.getsizeof(3340), "bytes")
print("Float:", sys.getsizeof(33.40), "bytes")
print("Decimal:", sys.getsizeof(Decimal("33.40")), "bytes")

# Time complexity
print("\nTIME COMPLEXITY")
print("All four methods take O(n) time for n payments.")
```